

$$\begin{aligned}
 E\left((y - \hat{f}(X))^2\right) &= \text{Bias}^2(\hat{f}(X)) + \text{Var}(\hat{f}(X)) + \sigma^2 \\
 &= \left(E(\hat{f}(X)) - E_{y|X}(f(X))\right)^2 + E\left((E(\hat{f}(X)) - \hat{f}(X))^2\right) + \sigma^2
 \end{aligned}$$